STEP CUTTING 4

AIM

To perform 'V' cutting on the given M.S. plate according to the given design.

Tools Required :

1. Bench vice
2. Try square
3. Hacksaw cutting
4. Scriber
5. Steel rule
6. Punches
7. Flat file – Rough
8. Flat file – medium
9. Flat file – smooth
10. Triangular file
11. Hammer
12. Divider
13. Chalk paste
14. Vernier caliper
15. Surface plate

PROCEDURE

1. The design of the given model is studied completely to understand the features like its Size and shape.
2. The given piece of metal is checked for sizes whether it is sufficient for the design.
3. Hand tools and measuring instruments are selected suitably to perform the required operations.
4. The piece of metal (Mild Steel) is held in the bench vice and two adjacent sides are filed for squareness using a flat file. These two sides are considered prime sides.
5. The remaining two sides are also filed for squareness(The angle between adjacent sides = 90 degrees). The same is checked with the help of a try – square.
6. The piece of work is checked for 90^0 on all four sides.
7. A thin layer of chalk paste is applied on the flat surface of the metal piece.
8. Chalk is allowed to dry.
9. The given design in the form of steps is clearly scribed on the flat chalked surface with the help of steel rule, divider, try square and scriber as per the given dimensions. The angular lines are scribed with the help of the protractor head of a combination set.
10. Punch marks are made at required points on the surface using a dot punch and a hammer.
11. Punch marks are made by keeping the punch inclined at 60^0so that they are filed off later.
12. The spacing between adjacent punch marks should be at least 6mm.
13. Thin auxiliary lines are made at a distance of 1.5mm from the punched line.
14. Hacksaw cuts are made on these auxiliary lines. The unwanted portion of the metal piece is removed.
15. The remaining portion of the metal piece is fitted on the bench vice. The rough surface resulting form hacksaw cutting is filed with the help of flat files (rough, medium and smooth)